

Mathematics Detailed Notes

WBJEE • JEE Main Mathematics preparation

1. Sets, Relations & Functions

- Understand set notation, subsets, union, intersection, complement and Venn-diagram based problems.
- Relations are described using ordered pairs; functions assign each input exactly one output.
- Revise domain, codomain and range carefully and practice composition and inverse-function questions.

2. Quadratic Equations & Complex Numbers

- Use the discriminant to classify real roots of a quadratic.
- Complex numbers are written as $a+ib$, where $i^2=-1$. Practice modulus, conjugate and basic algebra.
- Revise roots-coefficient relations and transformations of quadratic expressions.

3. Sequence & Series

- Arithmetic progression uses a constant common difference; geometric progression uses a constant common ratio.
- Practice n th-term and sum formulas and mixed problems involving AP/GP.
- Learn how to identify the progression before selecting a formula.

4. Permutation, Combination & Binomial Theorem

- Permutation focuses on arrangements; combination focuses on selections.
- Factorial notation and restrictions such as repetition, identical objects and fixed positions are common sources of mistakes.
- Binomial expansion gives coefficients through combinations; practice finding particular terms and coefficients.

5. Matrices & Determinants

- Revise matrix operations, transpose, determinant and inverse.
- For a 2×2 matrix, inverse exists when its determinant is non-zero.
- Use determinants for solving linear systems and geometry-style area questions.

6. Coordinate Geometry

- Straight lines: slope, intercept, distance and angle between lines are core skills.
- Circle problems commonly use centre-radius form and general equation transformations.
- Practice parabola, ellipse and hyperbola basics, standard forms and parameter interpretation.

7. Limits, Continuity & Differentiation

- Limits describe behavior near a point; continuity requires the limit, function value and appropriate agreement.
- Derivative represents instantaneous rate of change and slope.
- Practice algebraic simplification, standard limits and derivatives of common functions.

8. Application of Derivatives

- Use derivatives to study increasing/decreasing behavior, stationary points and maxima/minima.
- Critical points should be checked with an appropriate sign or second-derivative argument.
- Optimization questions require translating the word problem into a function first.

9. Integral Calculus

- Integration is the reverse process of differentiation and also represents accumulated quantity/area in suitable contexts.
- Revise standard integrals, substitution and integration by parts.
- Definite integrals use limits and can exploit symmetry and properties to simplify calculations.

10. Differential Equations

- Identify order and degree from the highest derivative present.
- Practice separation of variables and standard first-order forms.
- Always apply the given initial condition after obtaining the general solution when required.

11. Vector Algebra & 3D Geometry

- Vectors can be represented through components; magnitude and direction are fundamental.
- Dot product helps with angles and perpendicularity; cross product helps with area and perpendicular vectors.
- In 3D, revise direction ratios/cosines, lines, planes and distance relationships.

12. Probability & Statistics

- Probability questions often require counting the sample space correctly before applying probability rules.
- Revise conditional probability, independence, Bayes-type reasoning and random-variable basics.
- Statistics requires careful handling of mean, variance and standard deviation.

Revision Strategy

- Build speed by solving easy algebra first, then calculus/coordinate geometry, then lengthy mixed problems.
- Keep a separate error notebook for sign errors, domain restrictions, formula misuse and calculation mistakes.
- Do timed mixed sets regularly and revisit questions that took unusually long.